

Kartik Agnihotri

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Summary

Software Engineer with 2+ years of experience building production APIs, event-driven systems, and AI-powered backend platforms using Node.js, JavaScript, Redis, DynamoDB, PostgreSQL, and AWS. Experienced in multi-tenant architectures, asynchronous processing, LLM integrations, tool-calling, and agentic AI systems, with a strong track record of learning new technologies and taking ownership of unfamiliar technical problems end-to-end.

Experience

Full Stack Software Engineer, Fragmatic Technologies

April 2024 – Present

- Designed and owned **5 production REST APIs** for a multi-tenant B2B platform, handling authentication, validation, business logic, database integration, testing, and deployment; contributed to **40+ APIs** across the platform.
- Built **event-driven processing workflows** using **Redis, Bull, AWS SQS, and Lambda** for tracking workloads from websites with **~10K–1M monthly visitors**, implementing asynchronous jobs, retries, caching, and TTL-based processing.
- Built a **provider-agnostic AI service** integrating **OpenAI, Gemini, and Groq** through configurable provider adapters, allowing multiple product workflows to use a common LLM interface.
- Built an **agentic AI workflow in Node.js** for executing product operations through natural language, implementing **multi-step stateful orchestration, conditional routing, tool-calling loops, subagents, conversational context, authentication, logging, and failure handling** without relying on an agent orchestration framework.
- Built an **HTML personalization pipeline** for webpages of **~10K–35K characters**, recursively splitting HTML into model-sized components while preserving structure for deterministic reconstruction and validating generated output; extended the pipeline to support **image personalization**.

Associate Software Developer, OpenSense Labs

July 2023 – April 2024

- Implemented consent-aware tracking and customer segmentation workflows using **Apache Unomi**, automatically controlling data collection based on user consent and regional privacy requirements.
- Developed personalization and automation features across backend APIs and integrations, including a **RAG proof of concept** using **Node.js, OpenAI embeddings, Pinecone, and GPT-3.5-turbo**, with **chunking strategy design, embedding model selection, retrieval evaluation, and contextual retrieval** to improve the relevance and grounding of generated responses.

Projects

Distributed Job Scheduler | Node.js, Redis, PostgreSQL

- Built a distributed job scheduler from scratch with Redis-based queuing and PostgreSQL job state, supporting scheduled execution, retries with exponential backoff, worker coordination, and idempotent task processing.

KragDB | C++

Building a database system from scratch to explore storage, data management, and database internals.

Technical Skills

- **Languages:** JavaScript, Python, SQL, C++
- **Backend:** Node.js, Express.js, Fastify, REST APIs, Distributed Systems, Event-Driven Architecture, Worker Systems, Queue Processing, Async Pipelines
- **AI:** LLM Integration, Agentic AI, Tool/Function Calling, OpenAI, Gemini, Groq, RAG, Embeddings
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **Databases Caching:** DynamoDB, PostgreSQL, Redis, MongoDB
- **Cloud DevOps:** AWS Lambda, SQS, API Gateway, S3, IAM, VPC, CloudWatch, Docker

Education

Bachelor of Technology in Mechanical Engineering

July 2019 – June 2023

Ujjain Engineering College, Ujjain